

Nicholas Morello

✉ nicholas@vmello.dev | 🌐 vmello.dev | [in nvmello](https://www.linkedin.com/in/nvmello) | [github nvmello](https://github.com/nvmello)

SUMMARY

Software engineer focused on building systems that turn ideas into working software. Interested in how AI is changing the way we design, communicate, and ship. Currently building AI-assisted development tooling and workflows at Nextgear Capital, with a background in backend services and systems engineering.

EXPERIENCE

Nextgear Capital

July 2025 – Present

Backend Software Engineer

Indianapolis, IN

- Backend services for mobile and dealer-facing applications using Java and Python-based Azure Functions across a REST API microservices architecture
- Moved onto an **experimental team** within six months, focused on spec-driven development and establishing how AI-assisted workflows get built and adopted across the engineering org
- Self-initiated** and built a central AI skill hub with no team mandate behind it, now adopted by **150+ engineers** and maintained by **30+ contributors** org-wide
- Built **developer-facing tooling for context management, token usage monitoring, and external tool connection management**, plus shareable subagent scaffolding letting engineers reuse specialized AI agent configurations

Tech: Java, Python, Azure Functions, REST APIs, Microservices, Node.js, Claude Code, GitHub Copilot, Spec-Driven Development, Agentic Tooling

Raytheon Technologies

Feb. 2023 – July 2025

Embedded Software Engineer

Tucson, AZ

- Developed **embedded C++ software for mission-critical real-time systems**, ensuring real-time performance and reliability under strict aerospace standards
- Extensive **Python integration for firmware development**, working with hardware interfaces and automating validation workflows
- Optimized code for resource-constrained embedded platforms under aerospace certification requirements

Tech: C++, Python, RTOS, Embedded Systems

PROJECTS

Hue U | AI-Powered Coloring Book Platform

- Directed AI coding agents** to architect, implement, test, and ship this full-stack platform (web + native iOS) end-to-end, backed by full CI/CD and test coverage
- Engineered **AI photo-to-line-art generation** pipeline for converting uploaded images into printable coloring pages
- Built an interactive book builder with print-on-demand fulfillment and a **Stripe-powered credit system** for purchases
- Built a **native iOS app** (SwiftUI) with a custom flood-fill coloring engine and Apple Pencil support, currently in App Store review
- Built an **autonomous Hermes agent** for production operations, with approval-gated write actions behind a human-in-the-loop gate

Tech: Python, FastAPI, Next.js, Swift (iOS), Supabase, Google Gemini, Stripe, Docker, Google Cloud, Hermes, Agentic Development

vmello.dev | Portfolio & Live Dashboard

- Portfolio platform featuring **real-time fitness tracking**: a small Swift iOS companion app reads local HealthKit data and streams it to the dashboard over WebSocket
- Built a **Spotify and Apple Music integration** and a custom **animation system** (GSAP, Framer Motion) for the site

Tech: React, Node.js, Express.js, MongoDB, WebSocket, Swift (iOS), GSAP, Framer Motion, Tailwind CSS

EDUCATION

Arizona State University

Bachelor of Science in Computer Science

SKILLS

Languages : Java, C++, Python, Swift, JS / TS

Frameworks & Tools : React, Next.js, Python FastAPI, Azure, Supabase, MongoDB, Docker, REST APIs

AI & Methodology : Claude Code, Codex, Custom Pi Agent, Hermes Agent, Spec-Driven Development